

Jackson D. Maines

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Education

<i>Masters of Science</i> - Biostatistics University of Texas Health Science Center at Houston , Houston, TX	2025 - 2027
<i>Bachelors of Science</i> - Mathematics (3.97/4.0) University of Texas at Dallas, Richardson, TX	2023 - 2024
<i>Associate of Art</i> (3.3/4.0) Tarrant County College, Fort Worth, TX	2020 - 2023

Experience

Pacific Northwest National Lab — Richland, WA - <i>Masters Intern</i>	April 2025 – Present
• Aided in the development of a computer vision pipeline for automated segmentation of mutliwell culture plates. • Processed and segmented over 10,000 culture images, extracting key metrics including color, spatial coordinates, and size of inhibition zone. • Assisted researchers in applying bioinformatics tools and conducting data analyses.	
Pacific Northwest National Lab — Richland, WA - <i>Science Undergraduate Laboratory Internship</i>	January 2025 – April 2025
• Contributed to the synthetic biology groups investigation in the optimization of an antimicrobial compound. • Utilized Snakemake to analyze long-read PacBio sequencing data. • Leveraged Python and R to analyze optical density data, revealing variation that will be addressed in the recultivation of select samples. • Collaborated with Department of Energy scientist in the Earth and Biological Sciences and Energy and Environment directorates. • Wrote Report and Presented Poster: Optimization of Violacein Biosynthesis for Enhanced Antimicrobial Properties.	
University of Texas Medical Branch — Galveston, TX - <i>Summer Research Student</i>	June 2024 – July 2024
• Worked with infectious disease researcher analyzing data from SARS2 infected mice. • Collaborated with working biostatisticians to conduct analyzes in R. • Applied statistical analysis to RNA sequencing data to identify significant genes for future lab projects. • Analyzed significant genes in Ingenuity Pathway Analysis to explore gene pathways, results implied STING1 inhibition being linked to Xaf1 activation in MGL-1 KO SARS2 infected mice. • Presented poster: MGL-1: An Unrecognized Player in Xaf1 Pathway Activation Following SARS-CoV-2 Infection	

Skills

Programming Languages: R, MATLAB, Python, and LaTeX
Operating Systems: Linux/Unix system, Windows

Projects

Bike Sharing Analysis - <i>Group Project</i>
• Optimized a negative binomial regression model with variable selection and transformations.

Awards

Dean's List: UT-Dallas (Spring 2024, Spring 2023)